

A* Search Algorithm

A* (start, goal)

OPEN = {start}

while OPEN is not empty

current = node in OPEN with f(n)

if current == goal

return path to goal

else

remove current from OPEN

for each neighbour of current

calculate new path cost g(neighbour)

if this path is better

update parent of neighbour

update g(neighbour)

f(neighbour) = g(neighbour) + h(neighbour)

Return goal

2	8	3			1	2	3
1	6	4			8		4
	7	5			7	6	5

$$f(n) = g(n) + h(n)$$

g(n) → Depth of node

h(n) → Number of misplaced tile.

$$g(n) = 0$$

$$h(n) = 5$$

$$f(n) = 0 + 5 = 5$$

2	8	3
1	6	4
	7	5

$$g(n) = 1$$

2	8	3
	6	4
1	7	5

$$h(n) = 5$$

$$f(n) = 1 + 5$$

$$= 6$$



2	3	3
1	6	4
7		5

$$h(n) = 4$$

$$f(n) = 1 + 4$$

$$= 5$$



8 4
7 6 5

2	8	3
1	6	4
7		5

$g(n) = 2$

2	8	3
1		4
7	6	5

$h(n) = 3$

$f(n) = 5$ ✓

2	8	3
1	6	4
	7	5

$h(n) = 5$

$f(n) = 7$

2	8	3
1	6	4
7	5	

$h(n) = 5$

$f(n) = 7$

$g(n) = 3$

2		3
1	8	4
7	6	5

$h(n) = 3$

$f(n) = 6$ ✓

2	8	3
1	6	4
7		5

$h(n) = 4$

$f(n) = 7$

2	8	3
	1	4
7	6	5

$h(n) = 3$

$f(n) = 6$

2	8	3
	1	4
7	6	5

$f(n) = 4$

$f(n) = 7$

$g(n) = 4$

	2	3
1	8	4
7	6	5

$h(n) = 2$

$f(n) = 6$ ✓

2	3	
1	8	4
7	6	5

$h(n) = 4$

$f(n) = 8$

2	8	3
1		4
7	6	5

X

repeated

$g(n) = 5$

1	2	3
	8	4
7	6	5

$h(n) = 1$

$f(n) = 6$

2		3
1	8	4
7	6	5

$h(n) = 3$

$f(n) = 8$

$g(n) = 6$

1	2	3
8		4
7	6	5

$h(n) = 0$

$f(n) = 0 + 6$ Goal state.
= 6

	2	3
1	8	4
7	6	5

1	2	3
7	8	4
	6	5

$g(n)=3$

2	8	3
	1	4
7	6	5

$h(n)=3$ $f(n)=6$

$g(n)=4$

	8	3
2	1	4
7	6	5

2	8	3
1		4
7	6	5

2	8	3
7	1	4
	6	5

$h(n)=3$

$h(n)=3$

$h(n)=4$

$g(n)=5$ ✓

$g(n)=5$

2	8	3
	1	4
7	6	5

8		3
2	1	4
7	6	5

2		3
1	8	4
7	6	5

2	8	3
1	6	4
7		5

X

$h(n)=3$

$f(n)=8$

$f(n)=8$

$f(n)=9$

X repeated

$g(n)=6$

	8	3
2	1	4
7	6	5

8	3	
2	1	4
7	6	5

	2	3
1	8	4
7	6	5

2	3	
1	8	4
7	6	5

2	8	3
1		4
7	6	5

$h(n)=4$

$h(n)=3$

$h(n)=4$

$h(n)=3$

$f(n)=10$

$f(n)=9$

$f(n)=10$

$f(n)=9$

$f(n)=9$ ✓

$g(n)=7$

This will leads to maximum step
so previous one is the answer.

~~510~~

Exact